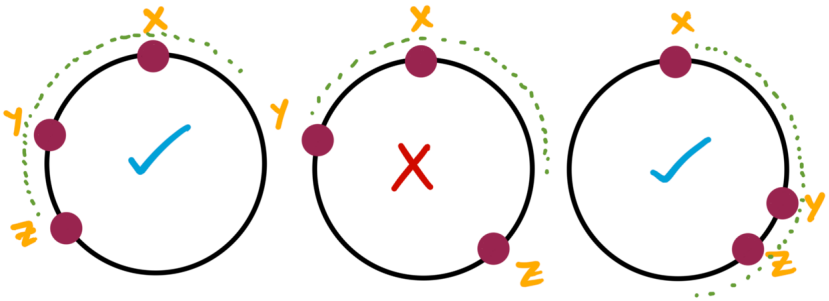
Wendel's Theorem

Ryan Peng

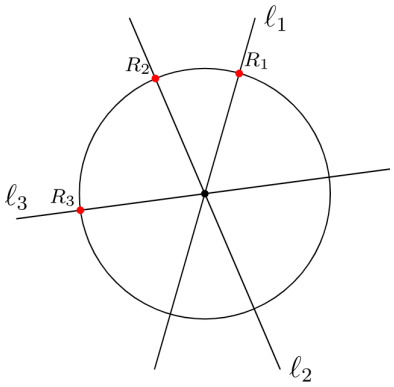
1 Three Points on a Circle

Suppose you have a circle, and you choose 3 random points on it. What is the probability that all 3 of these points lie on some half of the circle?

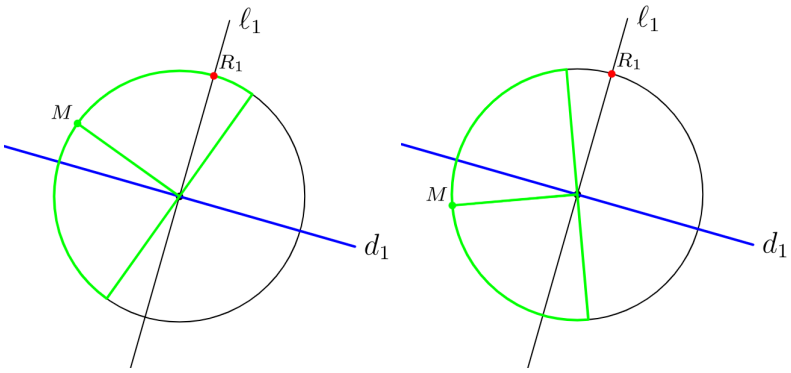


Here is one particular way to approach this problem:

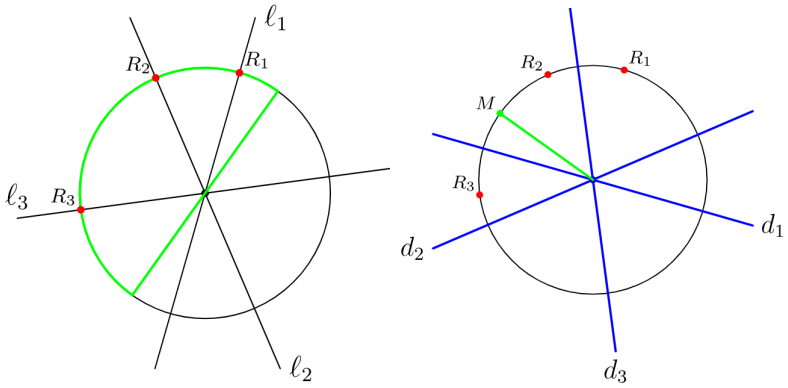
Label the three points R_1, R_2, R_3, and draw the lines connecting the points to the center of the circle. Call these lines ℓ_1, ℓ_2, ℓ_3, respectively. We can assume that ℓ_1, ℓ_2, ℓ_3 are distinct, since the probability that two randomly chosen points on a circle are diametrically opposite each other is 0. This is because there are infinitely many choices for the second point, but only one point that is diametrically opposite the first.



Consider a semicircle whose diameter does not coincide with any of the three lines. Let M be the midpoint of the semicircle.
Let d_1 be the line passing through the center of the circle that is perpendicular to ℓ_1.
Notice that the semicircle contains R_1 if and only if R_1 and M are on the same side of d_1.

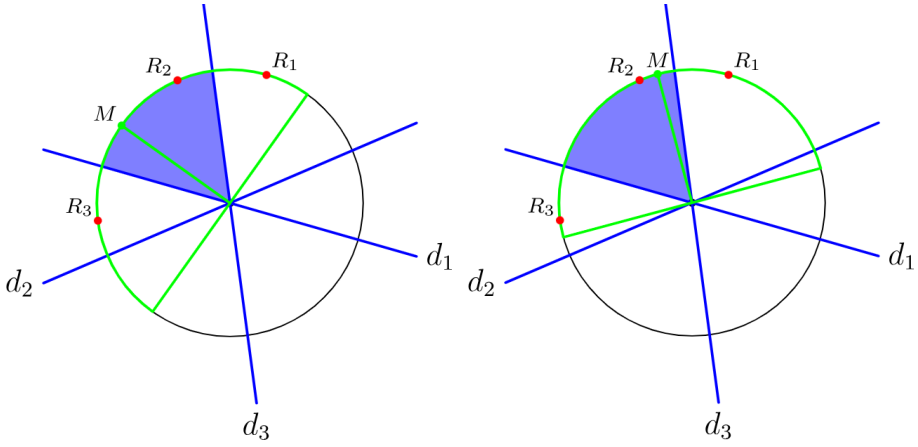


Define d_2 and d_3 analogously to d_1.



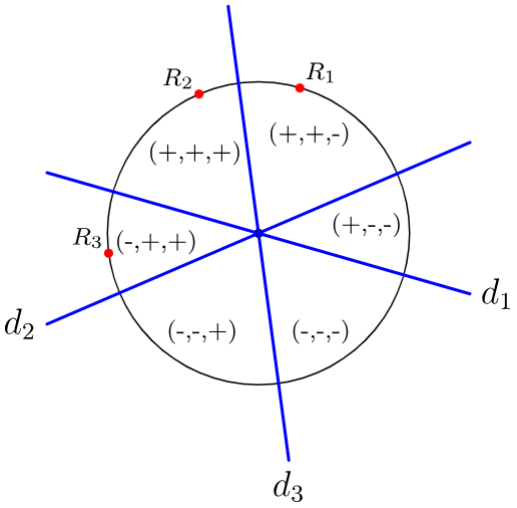
In this diagram, the semicircle with midpoint M contains all 3 of R_1, R_2, R_3. This is because M and R_1 are on the same side of d_1, M and R_2 are on the same side of d_2, and M and R_3 are on the same side of d_3.

Since this relies only on the position of M relative to d_1, d_2, d_3, the vector from M to the center of the circle can lie anywhere in the shaded region between d_1 and d_3, and the semicircle with midpoint M will still contain all 3 red points. However, if this vector lies outside of the shaded region, then the semicircle with midpoint M will not contain all 3 red points.

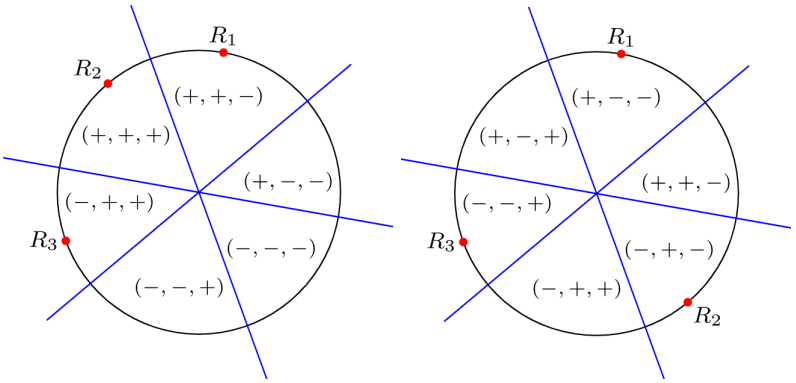


Extending this logic, each of the 6 regions that d_1, d_2, d_3 divide the plane into can be represented with a 3-tuple $s = (s_1, s_2, s_3)$, where $s_1, s_2, s_3 \in \{+, -\}$. For $1 \leq i \leq 3$, s_i is $+$ if and only if any semicircle with midpoint in the region contains R_i. Furthermore, we say that a 3-tuple (s_1, s_2, s_3) *occurs* if there exists some

region with $s = (s_1, s_2, s_3)$. For example, $(+,-,+)$ and $(-,+,-)$ do not occur in the diagram below. We want to find the probability that $(+,+,+)$ occurs.



Notice that reflecting R_i across the center of the circle will flip s_i in all 6 regions ($i = 2$ shown below).



Therefore, any sign is equally likely to be $+$ or $-$, so every 3-tuple is equally likely to occur.

Any 3 lines d_1, d_2, d_3 through the center of the circle divide the plane into 6 regions, and there are $2^3 = 8$ possible 3-tuples (s_1, s_2, s_3).

Therefore, the probability that $(+,+,+)$ occurs is $\frac{6}{8} = \boxed{\frac{3}{4}}$.

This might seem like an overly complicated solution for such a simple problem. And while it is true that simpler solutions exist, the strength in this approach is that it can be easily generalized, as we will see below.

2 More Points

What if instead of 3 points, we picked 4? 5? What about n points?

Well, notice that nowhere in the above argument did we rely on the number of points chosen being 3, except at the end when we calculated the numbers 6 and 8.

n lines will always divide the plane into $2n$ regions, and there are always 2^n possible $n-$tuples of $+$ and $-$

signs. Therefore, the general solution for n points is

$$P = \frac{\#\text{ of regions}}{\#\text{ of possible n-tuples}} = \frac{2n}{2^n} = \boxed{\frac{n}{2^{n-1}}}$$

3 Higher Dimensions

We can further generalize our result to higher dimensions. For example, instead of picking points on a circle, we can pick points on a sphere, and find the probability that they all lie on some hemisphere.

3.1 Definitions

To describe objects in higher dimensions, we must introduce some terms.

Definition 1

The **real coordinate space** of dimension d, denoted $\mathbb{R}^d$, is the set of all $d-$tuples of real numbers. For example, the real number line is $\mathbb{R}^1$ and the real coordinate plane is $\mathbb{R}^2$.

Definition 2

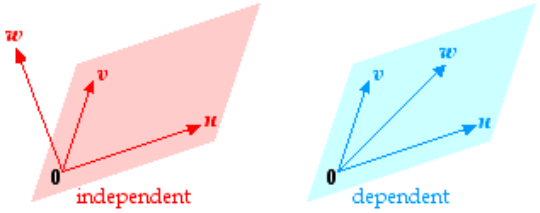
A **hyperplane** in $\mathbb{R}^d$ is a flat $(d-1)$-dimensional surface. It is represented by the linear equation

$$a_1x_1 + a_2x_2 + ... + a_dx_d = b$$

Where $a_1, a_2, ..., a_d$ are coefficients and b is a constant.
For example, a hyperplane in $\mathbb{R}^2$ is a line, and a hyperplane in $\mathbb{R}^3$ is a 2D plane.

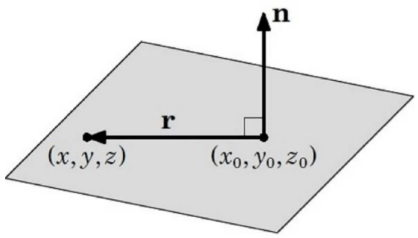
Definition 3

A set of vectors is said to be **linearly independent** if none of the vectors can be expressed as a linear combination of the others.



Definition 4

A **normal vector** to a hyperplane is a non-zero vector that is perpendicular to every vector lying within the hyperplane.



Definition 5

A set of hyperplanes is linearly independent if the set of their normal vectors is linearly independent.

Definitions 3, 4, and 5 are important because of the following fact:

Theorem 1

If a set of hyperplanes in $\mathbb{R}^d$ is linearly independent, then the intersection of any n of these hyperplanes has dimension $d - n$ if $n \leq d$, and is empty if $n > d$.

This is a consequence of the **rank-nullity theorem**, which states that the dimension of the solution set of a system of d independent linear equations in n variables is $d - n$.

Intuitively, $d - 1$ of the n variables can be eliminated by adding and subtracting multiples of the original equations. This leaves a linear equation in $d - n + 1$ variables, which describes a $(d - n)$-dimensional solution set.

$$4w + 7x + 3y + 5z = 12 \tag{1}$$
$$3w + 4x + 5y + 6z = 7 \tag{2}$$
$$6w + 10x + y + 3z = 4 \tag{3}$$

$$\text{Eliminiation: } 3 \cdot (1) - 2 \cdot (2) - 1 \cdot (3) \implies 3x - 2y = 18$$

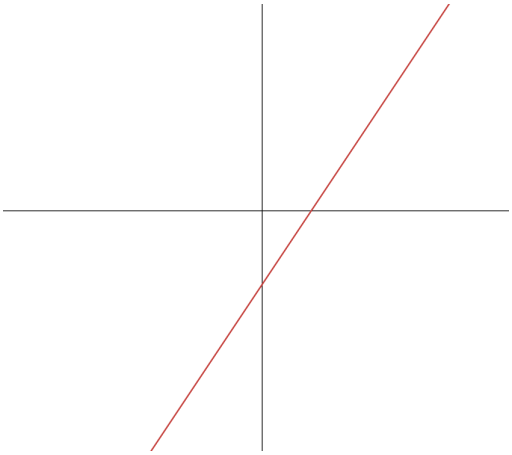


Figure 1: Solution set of 3 independent linear equations in 4 variables is a 1-dimensional line

3.2 Generalization

Now we can generalize our argument to higher dimensions.

Suppose we have a unit sphere centered at the origin in $\mathbb{R}^d$. Choose points $R_1, R_2, ..., R_n$ on the sphere at random.
For $1 \leq i \leq n$, let ℓ_1 be the line passing through R_i and the origin.
Then, let $d_1, d_2, ..., d_n$ be the hyperplanes through the origin perpendicular to $\ell_1, \ell_2, ..., \ell_n$, respectively. This means that $\overrightarrow{R_1}, \overrightarrow{R_2}, ..., \overrightarrow{R_n}$ are the normal vectors to $d_1, d_2, ..., d_n$.

We assume that any subset of size d of $\left\{\overrightarrow{R_1}, \overrightarrow{R_2}, ..., \overrightarrow{R_n}\right\}$ is linearly independent, therefore any subset of size d of $\{d_1, d_2, ..., d_n\}$ is linearly independent. We can do this because

- Since $d_1, d_2, ..., d_n$ all intersect at the origin, any subset of size greater than d of $\{d_1, d_2, ..., d_n\}$ cannot be linearly independent.
- We can make this assumption because the probability that d randomly chosen vectors in $\mathbb{R}^d$ are linearly independent is 1. Proving this rigorously requires measure theory, but intuitively this is true because any d linearly dependent vectors in $\mathbb{R}^d$ must all lie in some subspace of dimension less than d, and the probability that random vectors all lie in some lower-dimensional subspace is 0.

This is essentially a generalization of the assumption that no two points on the circle are diametrically opposite each other, from the original 2D problem.

Let $R(n, d)$ be the number of regions into which $d_1, d_2, ..., d_n$ divide $\mathbb{R}^d$. It turns out that $R(n, d)$ is constant for fixed n and d, given the above assumption. We will see why this is true in the next section.

Each region y can be assigned a $n-$tuple $s = (s_1, s_2, ..., s_n)$, where $s_i \in \{+, -\}$ for all $1 \leq i \leq n$. s_i is $+$ if a unit hemisphere centered at the origin with midpoint in y contains R_i, and $-$ otherwise. Here we define the midpoint of a hemisphere to be the intersection of the line through the center perpendicular to the base with the curved surface of the hemisphere.
Then $P(n, d)$ is equal to the probability that $(+, +, ..., +)$ occurs.
Since there are two choices for each s_i, there are 2^n possible $n-$tuples. Every $n-$tuple is equally likely to occur, and no two regions have the same $n-$tuple, so

$$\boxed{P(n, d) = \frac{R(n, d)}{2^n}}$$

4 Hyperplanes Through the Origin

In this section we will present and verify a general formula for $R(n, d)$.

4.1 Recurrence Relation

Suppose we have n hyperplanes through the origin in $\mathbb{R}^d$, whose normals are linearly independent in sets of d. These hyperplanes split $\mathbb{R}^d$ into $R(n, d)$ regions. Consider removing one of these hyperplanes, call it H_n. Then the number of regions that remain is $R(n-1, d)$. These regions can be split into two sets R_1 and R_2, such that for any region $r_i \in R_1$, $r_i \cap H_n$ is nonempty, and for any region $r_j \in R_2$, $r_j \cap H_n$ is empty.

$$R(n-1, d) = |R_1| + |R_2|$$

H_n splits each region in R_1 two and does not touch any region in R_2, so

$$R(n, d) = 2|R_1| + |R_2|$$

Substituting,

$$R(n, d) = R(n-1, d) + |R_1|$$

The intersection of each of the $n-1$ other hyperplanes with H_n is a hyperplane in the $(d-1)-$dimensional space H_n. Importantly, since we assumed that our original hyperplanes in $\mathbb{R}^d$ are linearly independent in sets of d, each of these intersections is distinct. Therefore, these intersections divide H_n into $R(n-1, d-1)$ regions. These regions are just the intersections of H_n with the regions in R_1, so

$$|R_1| = R(n, d-1)$$

Substituting, we obtain the recurrence relation

$$\boxed{R(n+1, d) = R(n, d) + R(n, d-1)}$$

4.2 Boundary Cases

Note that $R(n, d)$ must satisfy the following boundary cases:

- n lines through the origin divide the $2D$ plane into $2n$ regions, so $R(n, 2) = 2n$.
- Consider the case where $n \leq d$. We will show by induction that $R(n, d) = 2^n$.

  One hyperplane will always divide $\mathbb{R}^d$ in two, so $R(1, d) = 2$ for all d. This will be our base case. For the inductive step, assume that for some positive integer k, $k \leq d \implies R(k, d) = 2^k$ holds for all d. Then for all $d \geq k + 1$,

  $$R(k + 1, d) = R(k, d) + R(k, d - 1) = 2^k + 2^k = 2^{k+1}$$

  Since $k = 1$ satisfies $k \leq d \implies R(k, d) = 2^k$, by induction $n \leq d \implies R(n, d) = 2^n$ holds for all positive integers n.

4.3 The Formula

Consider the formula

$$R(n, d) = 2 \sum_{k=0}^{d-1} \binom{n-1}{k}$$

We can verify that this formula satisfies both boundary conditions:

$$\begin{aligned} R(n, 2) &= 2 \sum_{k=0}^{1} \binom{n-1}{k} \\ &= 2 \left(\binom{n-1}{0} + \binom{n-1}{1} \right) \\ &= 2(1 + (n - 1)) \\ &= 2n \end{aligned}$$

$\binom{n-1}{k} = 0$ if $k \geq n$, so if $n \leq d$,

$$\begin{aligned} R(n, d) &= 2 \sum_{k=0}^{d-1} \binom{n-1}{k} \\ &= 2 \sum_{k=0}^{n-1} \binom{n-1}{k} \\ &= 2 \left(2^{n-1} \right) \\ &= 2^n \end{aligned}$$

Lastly, verifying that this formula satisfies the recurrence relation:

$$\begin{aligned} R(n + 1, d) &= 2 \sum_{k=0}^{d-1} \binom{n}{k} \\ &= 2 \sum_{k=0}^{d-1} \left(\binom{n-1}{k} + \binom{n-1}{k-1} \right) \text{ (Pascal's identity)} \\ &= 2 \sum_{k=0}^{d-1} \binom{n-1}{k} + 2 \sum_{k=0}^{d-1} \binom{n-1}{k-1} \\ &= 2 \sum_{k=0}^{d-1} \binom{n-1}{k} + 2 \sum_{k=0}^{d-2} \binom{n-1}{k} \\ &= R(n, d) + R(n, d - 1) \end{aligned}$$

Theorem 2

n hyperplanes in $\mathbb{R}^d$ that each pass through the origin and are linearly independent in sets of d divide $\mathbb{R}^d$ into

$$2\sum_{k=0}^{d-1}\binom{n-1}{k}$$

disjoint regions.

This result is attributed to the Swiss mathematician Ludwig Schläfli.

5 Wendel's Theorem

Now that we have a formula for $R(n,d)$, we finally have the general solution

$$P(n,d) = \frac{R(n,d)}{2^n} = 2^{-n+1}\sum_{k=0}^{d-1}\binom{n-1}{k}$$

Theorem 3: Wendel's Theorem

The probability that n randomly chosen points on a sphere in $\mathbb{R}^d$ all lie on some hemisphere is

$$2^{-n+1}\sum_{k=0}^{d-1}\binom{n-1}{k}$$

This result was proven by J.G.Wendel in 1962.

6 Coin Tossing

Let's say you flip a fair coin $n-1$ times. What is the probability that you get less than d heads? Well, the probability of getting k heads in $n-1$ flips is

$$2^{-n+1}\binom{n-1}{k}$$

So the probability of getting less than d heads is

$$2^{-n+1}\sum_{k=0}^{d-1}\binom{n-1}{k}$$

Notice that this is exactly the same formula as in Wendel's Theorem!

In Wendel's original paper, he notes this connection, but remarks "it does not seem possible to find an isomorphism between coin-tossing and the given problem that would make the result immediate" [1].

While a direct mapping is elusive, we can draw parallels between coin tossing and the points-on-a-sphere problem.

First, we can convert the recurrence relation and boundary conditions on $R(n,d)$ into statements about $P(n,d)$.

Recurrence Relation:

$$R(n+1,d) = R(n,d) + R(n,d-1)$$

$$2^{n+1}P(n+1,d) = 2^nP(n,d) + 2^nP(n,d-1)$$

$$\boxed{P(n+1,d) = \frac{1}{2}P(n,d) + \frac{1}{2}P(n,d-1)}$$

Boundary Condition 1 ($d = 2$)

$$R(n,2) = 2n$$

$$2^nP(n,2) = 2n$$

$$\boxed{P(n,2) = \frac{n}{2^{n-1}}}$$

Boundary Condition 2 ($n \leq d$)

$$n \leq d \implies R(n,d) = 2^n$$

$$n \leq d \implies 2^nP(n,d) = 2^n$$

$$\boxed{n \leq d \implies P(n,d) = 1}$$

Let $C(n,d)$ be the probability of getting less than d heads in $n-1$ flips.

Suppose a coin is flipped n times and comes up heads less than d times. Then one of the following statements must be true:

- The coin came up heads less than d times in the first $n-1$ flips, then came up tails on the last flip.
- The coin came up heads less than $d-1$ times in the first $n-1$ flips, then came up heads on the last flip.

The last flip is equally likely to be heads or tails, so

$$C(n+1,d) = \frac{1}{2}C(n,d) + \frac{1}{2}C(n,d-1)$$

The probability of flipping zero heads in $n-1$ flips is $\frac{1}{2^{n-1}}$, and the probability of flipping heads exactly once is $\binom{n-1}{1} \cdot \frac{1}{2^{n-1}} = \frac{n-1}{2^{n-1}}$. Therefore, the probability of getting less than 2 heads in $n-1$ flips is $\frac{n}{2^{n-1}}$, so

$$C(n,2) = \frac{n}{2^{n-1}}$$

If $n \leq d$, then it is not possible to get more than $d-1$ heads in $n-1$ flips, so

$$n \leq d \implies C(n,d) = 1$$

Therefore, the boundary conditions and recurrence relation are the same, so

$$\boxed{P(n,d) = C(n,d)}$$

7 Conclusion

Wendel's Theorem is a fundamental result in geometric probability. While its statement might not seem practical, generalized versions of the theorem have surprising applications in various fields.

For example, forces on a particle can be modeled as points on a sphere, so Wendel's theorem can be applied to help calculate the local stability of particles.

Another example is in linear programming. Random linear programs are powerful optimization tools that take random vectors as parameters, and a generalized version of Wendel's Theorem can be used to calculate the probability that a random linear program is bounded.

Such applications commonly use the fact that if a set of points on a sphere do not all lie on a hemisphere, then their convex hull (the smallest possible convex shape that contains all of the points) contains the origin.

References

[1] Wendel, J. G. (1962). A Problem in Geometric Probability. *MATHEMATICA SCANDINAVICA, 11*, 109–112. https://doi.org/10.7146/math.scand.a-10655

[2] Sloughter, Dan. Lines, Planes, and Hyperplanes. LibreTexts Mathematics. https://math.libretexts.org/Bookshelves/Calculus/The_Calculus_of_Functions_of_Several_Variables_(Sloughter)/01%3A_Geometry_of_R/1.04%3A_Lines_Planes_and_Hyperplanes

[3] Thomas M. Cover. Bradley Efron. "Geometrical Probability and Random Points on a Hypersphere." Ann. Math. Statist. 38 (1) 213 - 220, February, 1967. https://doi.org/10.1214/aoms/1177699073

[4] Morse, P. K.; Corwin, E. I. Local Stability of Spheres via the Convex Hull and the Radical Voronoi Diagram. Phys. Rev. E 2023, 108 (6), 064901. https://doi.org/10.1103/PhysRevE.108.064901.

[5] Tikhomirov, K. (2025). On the probability that convex hull of random points contains the origin. https://arxiv.org/abs/2304.13133